$\sqrt{3}\ a = r_{Cs^+} + 2r_{Cl^-} + r_{Cs^+}$

$\left(\frac{\sqrt{3}}{2}\right) a = \left(r_{Cs^+} + r_{Cl^-}\right)$

$\left(\frac{\sqrt{3}}{2}\right) 400 =$ inter ionic distance

13. a) $\left(\frac{100}{0.414}\right)$

for an fcc structure $\frac{r_{X^+}}{r_{y^-}} = 0.414$

given that $r_{X^+} = 100\,pm$

$r_{y^-} = \frac{100\,pm}{0.414}$

14. c) 32%

packing efficiency = 68%

therefore empty space percentage = (100-68) = 32%

15. b) 848.5pm

let edge length =a

$\sqrt{2}a = 4r$

$a = \frac{4 \times 300}{\sqrt{2}}$

$a = 600 \times 1.414$

$a = 848.4$ pm

16. b) $\left(\frac{\pi}{6}\right)$

$\left(\frac{\frac{4}{3}\pi r^3}{a^3}\right) = \left(\frac{\frac{4}{3}\pi\left(\frac{a}{2}\right)^3}{a^3}\right) = \left(\frac{\pi}{6}\right)$

17. a) excitation of electrons in F centers

18. c) $\left(\frac{1}{2}a : \frac{\sqrt{3}}{4}a : \frac{1}{2\sqrt{2}}a\right)$

$sc \Rightarrow 2r = a \Rightarrow r = \frac{a}{2}$

$bcc \Rightarrow 4r = \sqrt{3}a \Rightarrow r = \frac{\sqrt{3}a}{4}$

$fcc \Rightarrow 4r = \sqrt{2}a \Rightarrow r = \frac{\sqrt{2}a}{4} = \frac{a}{2\sqrt{2}}$

$\left(\frac{a}{2}\right) : \left(\frac{\sqrt{3}a}{4}\right) : \left(\frac{a}{2\sqrt{2}}\right)$

19. d) $\left(\frac{\sqrt{3}}{2}\right)a$

if a is the length of the side, then the length of the leading diagonal passing through the body centered atom is $\sqrt{3}a$

Required distance $= \left(\frac{\sqrt{3}}{2}\right)a$

20. a) 915 kg m-3

$\rho = \frac{n \times M}{a^3 N_A}$

for bcc

$n = 2$

M=39

nearest distance 2r = 4.52

$a = \frac{4r}{\sqrt{3}} = \frac{2 \times 4.52 \times 10^{-10}}{\sqrt{3}} = 5.21 \times 10^{-10}$

$\rho = \frac{2 \times 39}{\left(5.21 \times 10^{-10}\right)^3 \times \left(6.023 \times 10^{23}\right)}$

$\rho = 915$ Kg m^{-3}

N_A

21. b) equal number of anions and anions are missing from the lattice

22. c) Frenkel defect

23. d) Both assertion and reason are false

24. b) FeO

25. a) XY_8

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